Moran's I and Geary's C

Mahedi Hasan and Jun Zhu

1 Moran's I and Geary's C

In this implementation, we use the small mosaic pattern II from lecture before to explore Moran's I and Geary's C.

1.1 Mosaic Pattern II

B	B	W	W
B	B	B	W
W	B	W	W

Recall that Moran's I statistic is a measure of spatial correlation and has the form

$$I = \frac{n}{\sum_i \sum_j w_{ij}} \cdot \frac{\sum_i \sum_j w_{ij}(X_i - \bar{X})(X_j - \bar{X})}{\sum_i (X_i - \bar{X})^2},$$

where n is the number of cells on the lattice, X_i is the variable of interest, $\bar{X}$ is the sample mean of X, and w_{ij} is the spatial weight between cell i and cell j.

Also recall that:

- if the spatial pattern is more regular, Moran's I would be negative and closer to -1;
- if the spatial pattern is more clustered, Moran's I would be positive and closer to $+1$;
- if the spatial pattern is complete spatial randomness (CSR), then Moran's I would be close to zero.

The function for Moran's I test in R is `moran.test()`, with the following syntax:

```
library(spdep)
args(moran.test)
```

```
## function (x, listw, randomisation = TRUE, zero.policy = attr(listw,
##     "zero.policy"), alternative = "greater", rank = FALSE, na.action = na.fail,
##     spChk = NULL, adjust.n = TRUE, drop.EI2 = FALSE)
## NULL
```

The default for this function sets `randomisation=TRUE`, which is used to estimate the variance of I and assumes that random samples were drawn under CSR. To estimate the variance by normal approximation, we set `randomisation=FALSE`, so we can reproduce the test result.

1.2 Data and spatial weights

We can apply Moran's I to the mosaic pattern II. It takes as input the data vector `egII` and a list of spatial weights `egII.lw`.

```
library(spdep)

egII <- c(1,1,0,0,
          1,1,1,0,
          0,1,0,0)

egII.nb <- cell2nb(nrow = 3, ncol = 4, type = "rook")
egII.lw <- nb2listw(egII.nb, style = "B")
```

2 Moran's I Test

We first compute Moran's I test under randomisation, which is the default.

```
set.seed(2)
moran.test(egII, egII.lw, alternative = "two.sided")
```

```
##
##  Moran I test under randomisation
##
## data:  egII
## weights: egII.lw
##
## Moran I statistic standard deviate = 1.1891, p-value = 0.2344
## alternative hypothesis: two.sided
## sample estimates:
## Moran I statistic       Expectation          Variance
##        0.17647059       -0.09090909        0.05055907
```

The observed Moran's I statistic is approximately 0.21. The standard deviate (or z-value) is approximately 1.2811, and the p-value for this two-sided test is 0.2002. We do not reject the null of CSR at the 10% level. There is no evidence of departure from CSR.

The default Moran's I test uses Monte Carlo or randomisation-based variance. Alternatively, a normal approximation can be applied by setting `randomisation=FALSE` in the function.

```
moran.test(egII, egII.lw, alternative = "two.sided", randomisation = FALSE)
```

```
##
##  Moran I test under normality
##
## data:  egII
## weights: egII.lw
##
## Moran I statistic standard deviate = 1.2996, p-value = 0.1938
## alternative hypothesis: two.sided
## sample estimates:
```

```
## Moran I statistic       Expectation          Variance
##        0.17647059       -0.09090909        0.04233200
```

Under `randomisation = FALSE` versus `randomisation = TRUE` (the default), the variances are different. Under randomisation, the variance is about `0.04546288`, but under normality, it is about `0.04541334`. The p-value under normal approximation is about `0.1603`, smaller than the previous p-value under randomisation. Under the normal approximation, we still do not reject the null of CSR at the 10% level. There is no evidence against CSR, supporting a non-random spatial pattern only weakly.

3 Geary's C Test

Recall that Geary's C is a measure of dissimilarity and has the form

$$C = \frac{(n-1)\sum_i \sum_j w_{ij}(X_i - X_j)^2}{2\left(\sum_i \sum_j w_{ij}\right)\sum_i (X_i - \bar{X})^2}.$$

Geary's C ranges from 0 to 2. A C-value around 1 yields no evidence of spatial autocorrelation. A value less than 1 suggests positive spatial correlation, whereas a value greater than 1 indicates negative spatial correlation.

The function in R to perform Geary's test is `geary.test()`, with the following syntax:

```
args(geary.test)
```

```
## function (x, listw, randomisation = TRUE, zero.policy = attr(listw,
##     "zero.policy"), alternative = "greater", spChk = NULL, adjust.n = TRUE)
## NULL
```

Similar to `moran.test()`, the default is `randomisation=TRUE`.

We now apply `geary.test()` to the mosaic pattern II and again set the seed value as before.

```
set.seed(3)
geary.test(egII, egII.lw, alternative = "two.sided")
```

```
##
##  Geary C test under randomisation
##
## data:  egII
## weights: egII.lw
##
## Geary C statistic standard deviate = 1.1891, p-value = 0.2344
## alternative hypothesis: two.sided
## sample estimates:
## Geary C statistic       Expectation          Variance
##        0.75490196        1.00000000        0.04248366
```

We know that under CSR, the expectation of the test statistic C is 1. The observed test statistic is about 0.73, the standard deviate (or z-value) is about 1.2811, and the p-value is 0.2002. We do not reject the null hypothesis of CSR at the 10% level. There is no evidence against CSR.

Similar to `moran.test()`, we can set `randomisation=FALSE` to calculate the variance under a normal approximation.

```
geary.test(egII, egII.lw, alternative = "two.sided", randomisation = FALSE)
```

```
##
##  Geary C test under normality
##
## data:  egII
## weights: egII.lw
##
## Geary C statistic standard deviate = 1.1522, p-value = 0.2492
## alternative hypothesis: two.sided
## sample estimates:
## Geary C statistic       Expectation          Variance
##        0.75490196         1.00000000        0.04524887
```

The variance under normality is approximately `0.04063539`, slightly different from the variance under randomisation, which is approximately `0.04584798`. The z-value is approximately `1.3608`, compared to `1.2811` under randomisation. Consequently, the p-value is also different at `0.1736`, smaller than the previous p-value under randomisation. Under the normal approximation, we still do not reject the null of CSR at the 10% level. There is no evidence against CSR.

4 Moran's I Test by Monte Carlo

Moran's I and Geary's C tests above are based on a z-value, and the variance is estimated by Monte Carlo or normal approximations. Next, we use a Monte Carlo method to directly get the p-values by considering repeated values to cells $nsim$ times under the null hypothesis of CSR.

The function in R to perform Moran's I test by Monte Carlo is `moran.mc()`.

We start with 99 Monte Carlo simulations, or `nsim = 99`.

```
set.seed(4)
moran.mc(egII, egII.lw, nsim = 99)
```

```
##
##  Monte-Carlo simulation of Moran I
##
## data:  egII
## weights: egII.lw
## number of simulations + 1: 100
##
## statistic = 0.17647, observed rank = 91, p-value = 0.09
## alternative hypothesis: greater
```

From the output, the test statistic is approximately `0.20833`, and the p-value for the two-sided test is `0.08` times 2, or `0.16`. We do not reject the null at the 10% level. There is no evidence against CSR.

Next, we change `nsim` to 999.

```
set.seed(6)
moran.mc(egII, egII.lw, nsim = 999)
```

```
##
##  Monte-Carlo simulation of Moran I
##
## data:  egII
## weights: egII.lw
## number of simulations + 1: 1000
##
## statistic = 0.17647, observed rank = 876, p-value = 0.124
## alternative hypothesis: greater
```

Here, the test statistic remains the same and the p-value changes slightly. The results are essentially unchanged, and we still do not reject the null at the 10% level. There is still no evidence against CSR.

5 Geary's C by Monte Carlo

Similar to Moran's I, there is also a Monte Carlo method for Geary's C, with the function `geary.mc()`.

We use the R function `geary.mc()`, which takes the arguments of the data vector `egII` and the list of spatial weights `egII.lw`. We first set `nsim` equal to 99.

```
set.seed(7)
geary.mc(egII, egII.lw, nsim = 99)
```

```
##
##  Monte-Carlo simulation of Geary C
##
## data:  egII
## weights: egII.lw
## number of simulations + 1: 100
##
## statistic = 0.7549, observed rank = 10, p-value = 0.1
## alternative hypothesis: greater
```

The output shows that the test statistic c is about `0.7257`. The p-value for the two-sided test is 2 times `0.095`, or `0.19`, indicating that we do not reject the null hypothesis of CSR at the 10% level. There is no evidence against CSR.

Now increase `nsim` up to 999.

```
set.seed(8)
geary.mc(egII, egII.lw, nsim = 999)
```

```
##
##  Monte-Carlo simulation of Geary C
##
## data:  egII
```

```
## weights: egII.lw
## number of simulations + 1: 1000
##
## statistic = 0.7549, observed rank = 120, p-value = 0.12
## alternative hypothesis: greater
```

The test statistic remains the same across both tests at approximately `0.7257`. The p-value changes slightly up to 2 times `0.0885`, or `0.177`, when `nsim` goes up to 999. We do not reject the null hypothesis of CSR at the 10% level. There is no evidence against CSR.

6 Summary

Using Moran's I and Geary's C, under both randomisation-based and normal-approximation variance calculations, and also under Monte Carlo testing, we do not reject the null hypothesis of CSR for mosaic pattern II at the 10% level.

The results consistently indicate that there is no strong evidence against CSR in this example.